



Forward Modeling the Galaxy Mass—Metallicity Relation

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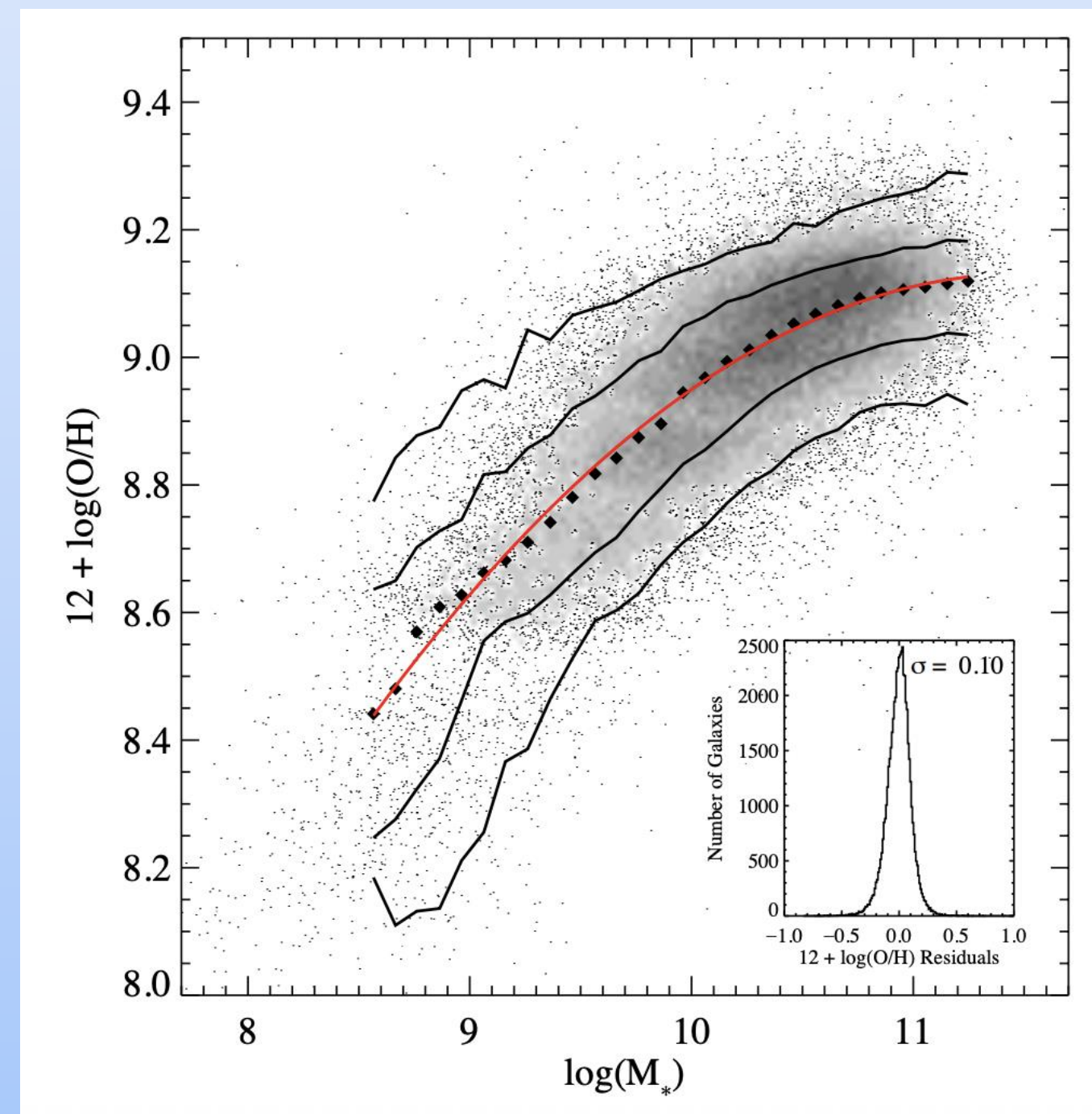


The Mass—Metallicity Relation

The mass-metallicity relation (MZR) describes the **tight empirical correlation between the stellar masses of a galaxy population and those galaxies' metallicities**, both increased as generations of stars live and die.

- The MZR is sensitive to:
- Star formation
 - Stellar and galactic winds
 - Supermassive black hole activity
 - Gas inflows and outflows
 - More!

Characterizing the evolution of this relation across the Universe's history is thus of critical importance for **understanding the evolution of the entire galaxy population**.



An example mass-metallicity relation (MZR) for a population of galaxies in the local Universe from Tremonti et al. (2004).

Key Terms

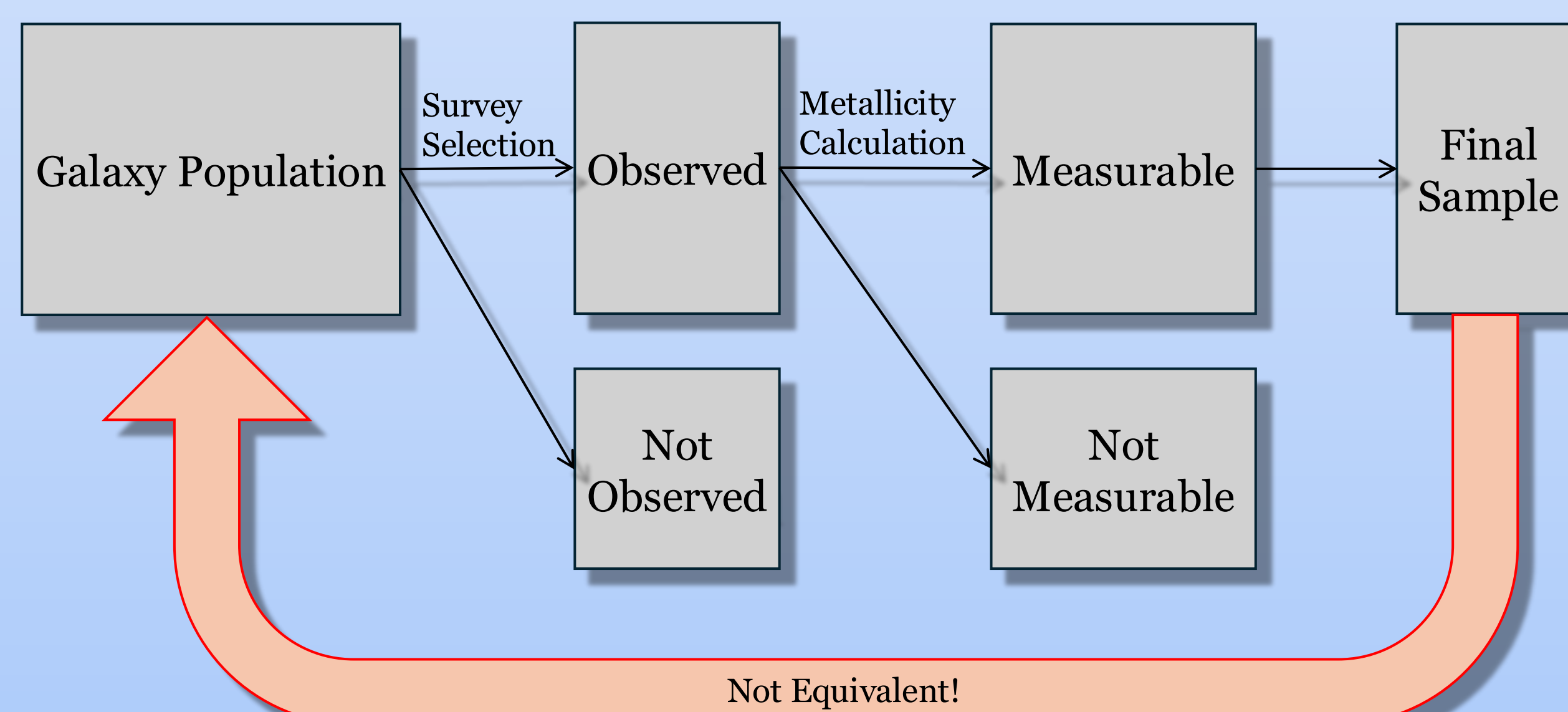
Stellar mass: amount of mass in stars in a galaxy

Metallicity: Also known as "chemical enrichment", describes the amount of "metals" (any element heavier than helium) relative to hydrogen in a galaxy

Survey Biases and the MZR

A typically unstated requirement of placing a galaxy on the mass-metallicity plane is that both its stellar mass and metallicity can be measured. Two stipulations must be true for this to happen: **first, the galaxy must be observed**.

Galaxy surveys are not always designed with metallicity studies in mind. Galaxies are often targeted based on separate physical characteristics that correlate with mass and metallicity, such as luminosity, effective color, or environment.

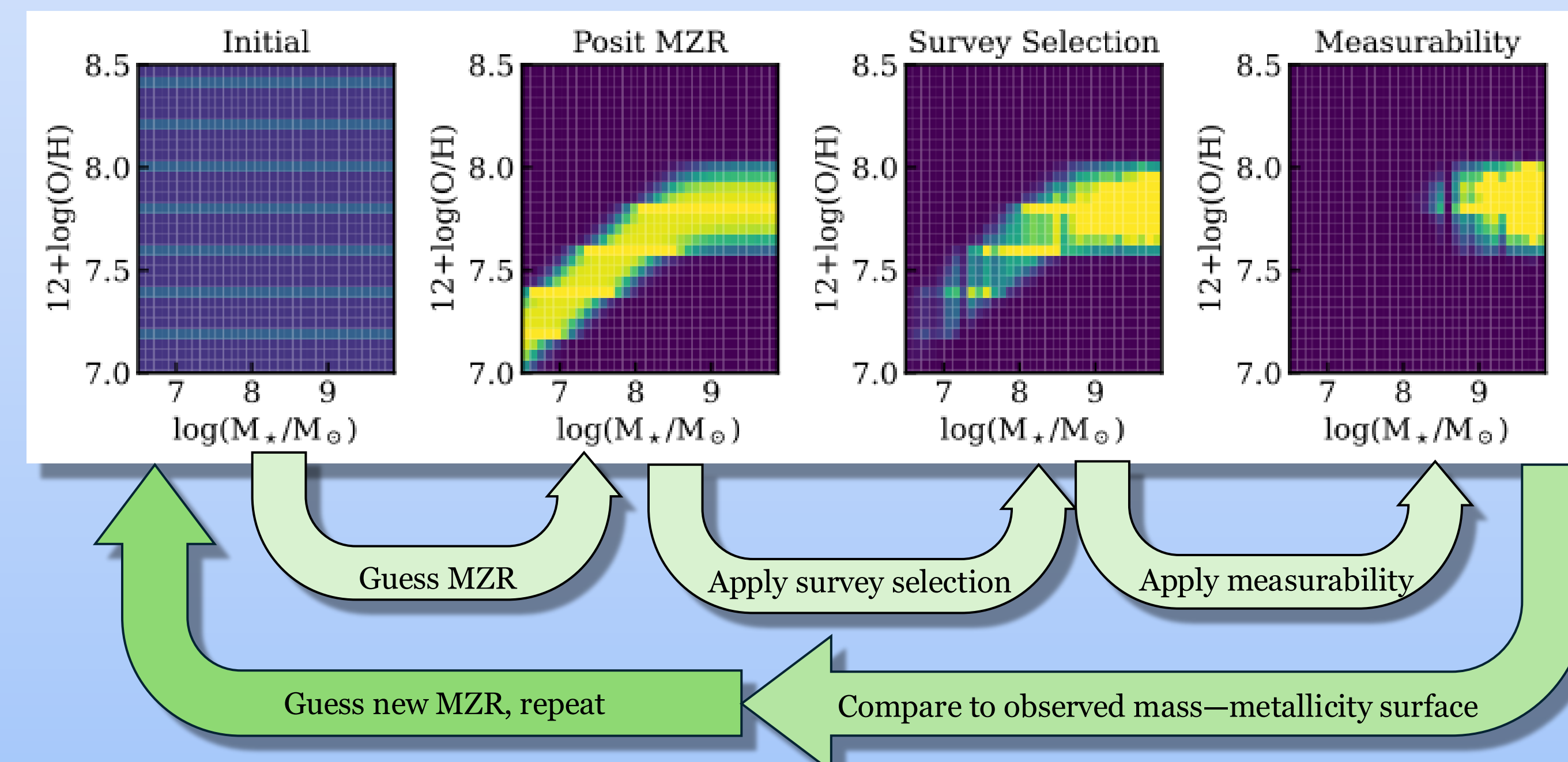


Second, **the galaxy must have a measurable metallicity**. This most often means that the galaxy must have a star formation rate high enough that its emission lines are able to be measured and compared to known relationships between chemical enrichment and emission line flux ratios. Galaxies with low amounts of star formation will be too dim to characterize.

These two aspects mean that **an observed population of galaxies is not usually representative of the overall galaxy population**. This leads to bias introduced into measured MZR, skewing our understanding of galaxy evolution.

Can We Better Recover an MZR?

We develop a framework to forward model the MZR using our knowledge of these biases. We use the *prospector* code^[2] to generate artificial galaxy models, varying their stellar mass, metallicity, star formation rate, and redshift, and assigning properties such as dust content and star formation history accordingly.



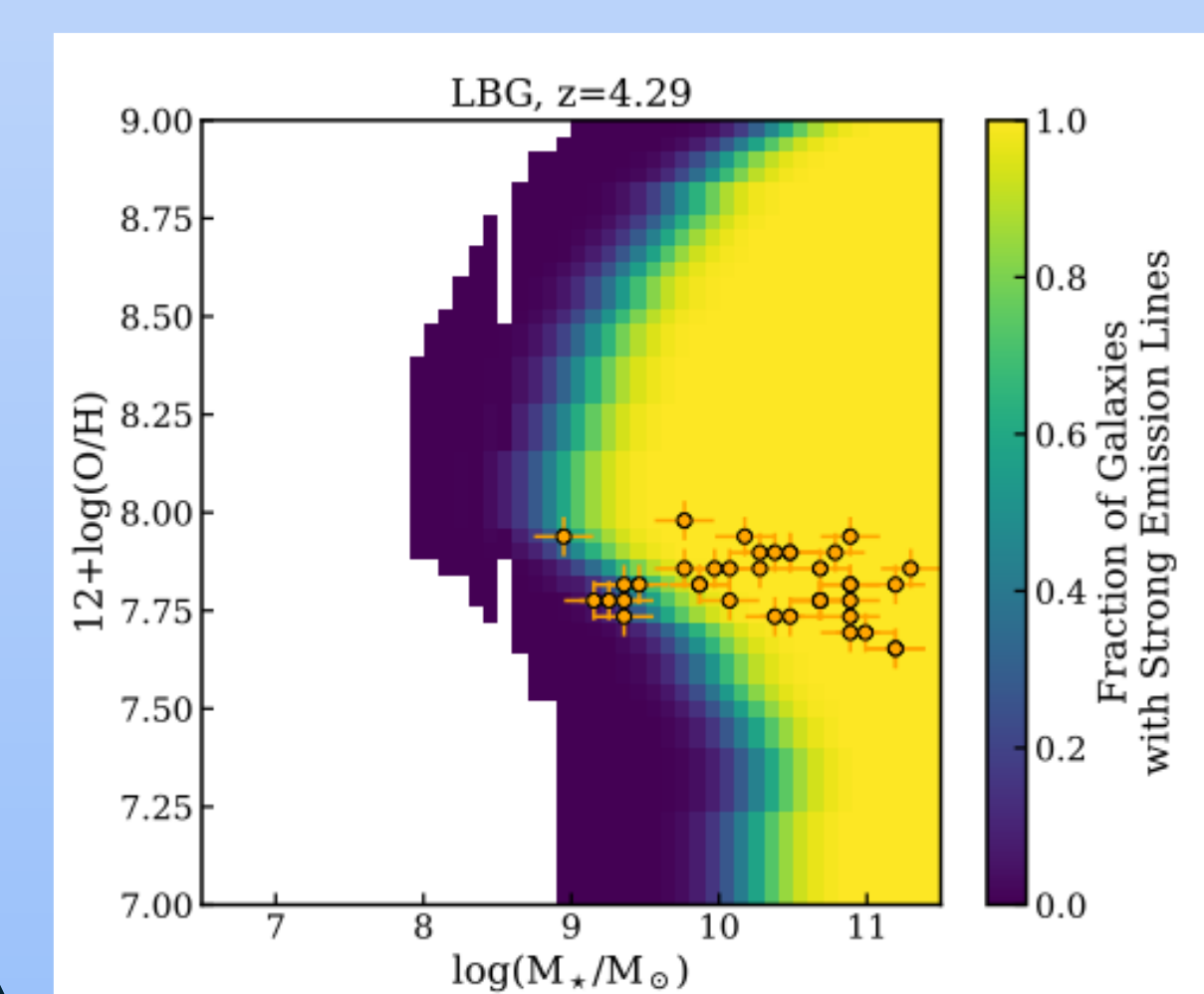
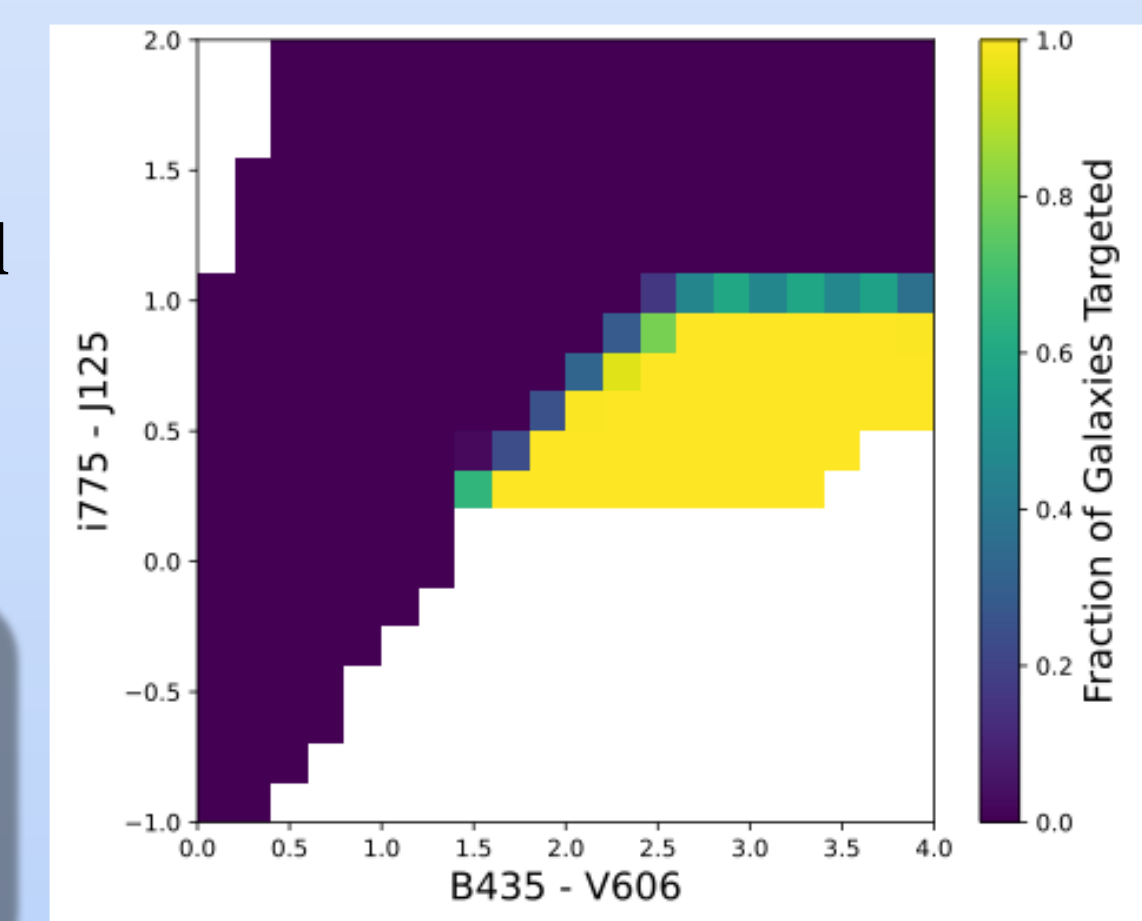
The forward modeling process. First, we posit an MZR. Next, we remove galaxies that a given survey would not have observed. Finally, we remove galaxies with emission lines too weak to characterize, leaving a mass-metallicity surface we can match to the observed surface. We iterate through this process until converging on an input MZR that, when processed, is congruent with the observed MZR.

Mock Survey Test

To test this process, we create a mock survey with a known input MZR and attempt to recover it. Our mock survey uses a color-color cut from Bouwens et al. (2015) to select Lyman-break galaxies (LBGs) at $z \sim 4$.

We first model the LBG selection function in color-color space using *prospector*. For each cell in parameter space, we measure the fraction of galaxies that would have been targeted by this mock survey.

Important: color is correlated with mass and metallicity. Failing to take this survey selection into account will result in incorrect conclusions about the MZR!

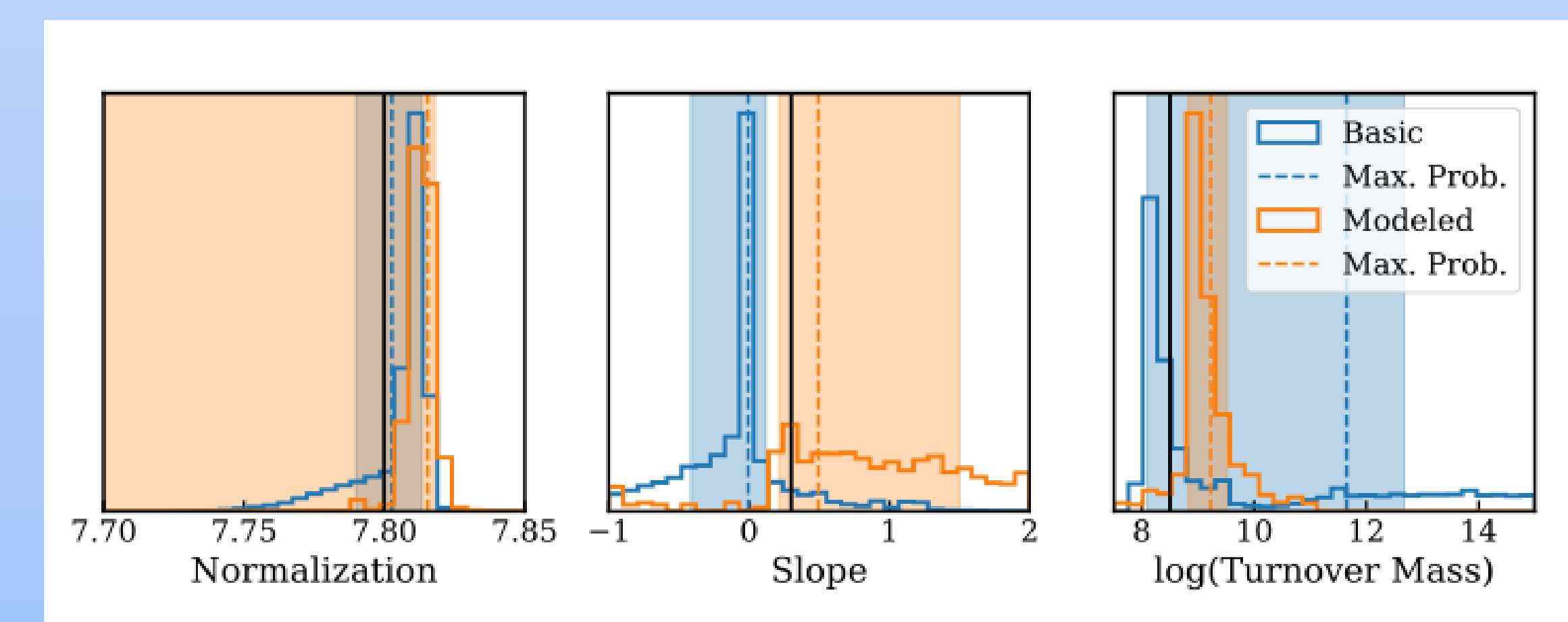
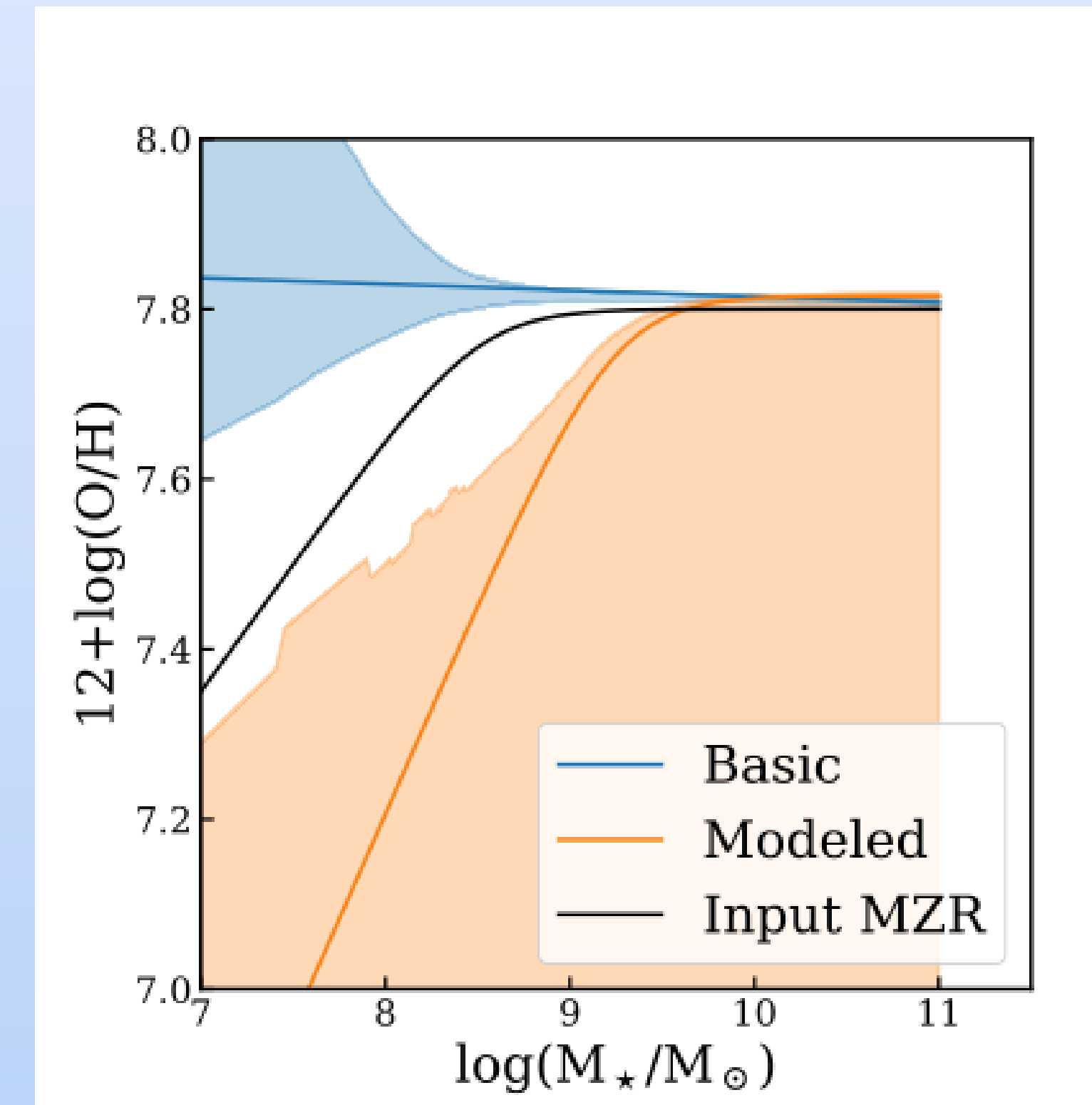


We then model emission line measurability using *prospector*. For each cell in mass and metallicity space in a slice of cosmic time, we measure the fraction of galaxies that have emission lines strong enough to be measured at that distance.

Important: star formation rate is correlated with mass and metallicity. Failing to take this measurability into account will result in incorrect conclusions about the MZR!

Results

We find that **performing a "basic", non-modeled fit to the observed MZR leads to qualitatively different results than when forward modeling**. While neither method perfectly captures the input MZR, only the forward modeled version reconstructs the "break" in the MZR as opposed to fitting a flat relation. The modeling also helps better estimate parameter uncertainties. This test verifies that this framework is a worthwhile and important investment for metallicity studies.



Basic: converges to correct normalization

Basic: converges to flat slope!

Basic: cannot converge to turnover mass

Modeled: converges to correct normalization

Modeled: converges to positive, but high, slope

Modeled: converges to slightly high turnover mass

Future Work

The next step of this process is to **apply the forward modeling framework to real observed metallicity studies**. By better estimating MZR parameters and uncertainties across a range of cosmic times, we will be able to constrain the chemical evolution of galaxies in a more robust way than ever before, significantly enhancing our knowledge of how galaxies change over time.

Acknowledgments and References

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- [1] Tremonti, C. A., Heckman, T. M., Kauffmann, G., et al. 2004, ApJ, 613, 898
- [2] Johnson, B. D., Leja, J., Conroy, C., & Speagle, J. S. 2021, 971 ApJS, 254, 22
- [3] Bouwens, R. J., Illingworth, G. D., Oesch, P. A., et al. 2015, ApJ, 803, 34